

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

__show ip route ospf__

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? __S0/0/0__

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? __129__

Does R2 show up as an OSPF neighbor on R1? __Yes__

Does R2 show up as an OSPF neighbor on R3? __No__

What does this information tell you?

__R2 only has OSPF active on S0/0/0, so it forms adjacency only with R1. R3 has no adjacency with R2, so R3 reaches 192.168.2.0/24 via R1.

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

_R2(config)# router ospf 1__

_R2(config-router)# no passive-interface ~~S0/0/0~~ S0/0/1__

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? __S0/0/1__

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

__T1 serial link = 64 (cost)

__R2 G0/0 = 1 (cost)

64+1=65_____

Is R2 listed as an OSPF neighbor to R3? ___Yes___

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

___To distinguish costs on links faster than 100 Mb/s. Otherwise FastE/GigE/10G may all have cost._____

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

___192.168.3.0/24: R1 S0/0/1 + R3 G0/0 (781+1=782)_____

___192.168.23.0/30: R1 S0/0/1 + R3 S0/0/1 (781+64=845)_____

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

___1562. Because each 128 Kb/s serial interface has cost about 781, so $781 + 781 = 1562$._____

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Because OSPF selects the lower-cost path. The cost on R1 S0/0/1 was increased, making the path via R3 higher cost than via R2._____

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

Router ID affects DR/BDR elections on multi-access networks and is the unique identifier in OSPF. If the ID is automatically chosen from a physical interface address, it can change after interface events and may disrupt adjacencies. Controlling router ID assignment keeps RID stable and predictable.

2. Why is the DR/BDR election process not a concern in this lab?

DR/BDR is mainly a feature of multi-access networks where many routers share the same segment. This lab uses point-to-point serial connections between routers, so there is no shared multi-access segment on those links and no DR/BDR election occurs, making it irrelevant here.

3. Why would you want to set an OSPF interface to passive?

A passive OSPF interface prevents neighbor formation and stops OSPF hello traffic on that interface, which reduces unnecessary protocol overhead and limits exposure to unwanted adjacency attempts. The connected subnet can still be advertised into OSPF, so other routers can reach it without running OSPF with end devices on that LAN.

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

Area-based hierarchy (Area 0 backbone + other areas), enabling LSA containment and route summarization.

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

network 10.0.0.0 0.255.255.255 area 0

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

EIGRP, because it has the lowest administrative distance, which is 90.

